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1 Introduction and the Kaluza–Klein Hypothesis

The primary source for these notes is [1].

The motivation for Kaluza–Klein theory is to attempt a unification of electrodynamics with general relativity. We recall that in electrodynamics we have the four-potential A_μ given by

$$A = -\phi dt + A_1 dx + A_2 dy + A_3 dz$$

where ϕ is the electric potential and $\vec{A} = (A_1, A_2, A_3)$ is the magnetic potential (so $\vec{E} = -\nabla\phi$ and $\vec{H} = \nabla \times \vec{A}$). Through the four-potential we can define the electromagnetic tensor

$$F_{\mu\nu} = \frac{\partial}{\partial x^\mu} A_\nu - \frac{\partial}{\partial x^\nu} A_\mu$$

which has coefficients

$$[F_{\mu\nu}] = \begin{bmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & H_3 & -H_2 \\ E_2 & -H_3 & 0 & H_1 \\ E_3 & H_2 & -H_1 & 0 \end{bmatrix}.$$

Just as A_μ acts as the electric potential, in general relativity we have the metric $g_{\mu\nu}$ which acts as the “gravitational potential.” The first step in the attempt to unify the two areas of physics is to combine the potentials into a common object. To do so, we begin with a four-dimensional spacetime with coordinates $(x^0, x^1, x^2, x^3) = (t, x, y, z)$ which has a metric $g_{\mu\nu}$, an electromagnetic four-potential A_μ , and a scalar field ψ . We then consider an extension to a five-dimensional spacetime with the new coordinate $x^5 = w$. As convention, we skip the index 4 and use 5 to denote the extra new dimension; we will use capital letters (i.e. M or N) for 5-dimensional indices and greek letters (i.e. μ or ν) for the indices 0, 1, 2, 3. We assume the metric on this space h has the following structure which extends the structure on the four-dimensional space:

$$h = (g_{\mu\nu} + \psi^2 A_\mu A_\nu) dx^\mu dx^\nu + \psi^2 A_\mu dx^5 dx^\mu + \psi^2 A_\nu dx^\nu dx^5 + \psi^2 (dx^5)^2.$$

From this we can see that the dual metric is given by

$$g^{\mu\nu} \frac{\partial}{\partial x^\mu} \frac{\partial}{\partial x^\nu} - A^\mu \frac{\partial}{\partial x^\mu} \frac{\partial}{\partial x^5} - A^\nu \frac{\partial}{\partial x^5} \frac{\partial}{\partial x^\nu} + \left(A^\sigma A_\sigma + \frac{1}{\psi^2} \right) \left(\frac{\partial}{\partial x^5} \right)^2.$$

2 The Kaluza–Klein Miracle

As with other metric forms (such as Schwarzschild or Robertson-Walker), we can attempt to write out the Ricci curvature tensor, the Ricci scalar curvature, and perhaps even the Einstein Field equations. We first recall that the Christoffel symbols are given by

$$\hat{\Gamma}_{NL}^M = \frac{1}{2} h^{MK} \left(\frac{\partial}{\partial x^N} h_{LK} + \frac{\partial}{\partial x^L} h_{NK} - \frac{\partial}{\partial x^K} h_{NL} \right).$$

By assumption, there is no w -dependence on $g_{\mu\nu}$, A_μ , or ψ . This immediately gives that

$$\begin{aligned} \hat{\Gamma}_{\nu\lambda}^\mu &= \frac{1}{2} h^{\mu K} \left(\frac{\partial}{\partial x^\nu} h_{\lambda K} + \frac{\partial}{\partial x^\lambda} h_{\nu K} - \frac{\partial}{\partial x^K} h_{\nu\lambda} \right) \\ &= \frac{1}{2} h^{\mu\sigma} \left(\frac{\partial}{\partial x^\nu} h_{\lambda\sigma} + \frac{\partial}{\partial x^\lambda} h_{\nu\sigma} - \frac{\partial}{\partial x^\sigma} h_{\nu\lambda} \right) + \frac{1}{2} h^{\mu 5} \left(\frac{\partial}{\partial x^\nu} h_{\lambda 5} + \frac{\partial}{\partial x^\lambda} h_{\nu 5} - \frac{\partial}{\partial x^5} h_{\nu\lambda} \right) \\ &= \frac{1}{2} g^{\mu\sigma} \left(\frac{\partial}{\partial x^\nu} (g_{\lambda\sigma} + \psi^2 A_\lambda A_\sigma) + \frac{\partial}{\partial x^\lambda} (g_{\nu\sigma} + \psi^2 A_\nu A_\sigma) - \frac{\partial}{\partial x^\sigma} (g_{\nu\lambda} + \psi^2 A_\nu A_\lambda) \right) - \frac{1}{2} A^\mu \left(\frac{\partial}{\partial x^\nu} (\psi^2 A_\lambda) + \frac{\partial}{\partial x^\lambda} (\psi^2 A_\nu) \right) \\ &= \Gamma_{\nu\lambda}^\mu + \frac{1}{2} g^{\mu\sigma} \left(\frac{\partial}{\partial x^\nu} (\psi^2 A_\lambda A_\sigma) + \frac{\partial}{\partial x^\lambda} (\psi^2 A_\nu A_\sigma) - \frac{\partial}{\partial x^\sigma} (\psi^2 A_\nu A_\lambda) \right) - \frac{1}{2} A^\mu \left(\frac{\partial}{\partial x^\nu} (\psi^2 A_\lambda) + \frac{\partial}{\partial x^\lambda} (\psi^2 A_\nu) \right). \end{aligned}$$

Expanding the second term with the Leibniz rule, we obtain

$$\begin{aligned} &\frac{1}{2} g^{\mu\sigma} \left(\frac{\partial}{\partial x^\nu} (\psi^2 A_\lambda A_\sigma) + \frac{\partial}{\partial x^\lambda} (\psi^2 A_\nu A_\sigma) - \frac{\partial}{\partial x^\sigma} (\psi^2 A_\nu A_\lambda) \right) \\ &= \frac{1}{2} g^{\mu\sigma} \left(\frac{\partial(\psi^2 A_\lambda)}{\partial x^\nu} A_\sigma + \psi^2 A_\lambda \frac{\partial A_\sigma}{\partial x^\nu} + \frac{\partial(\psi^2 A_\nu)}{\partial x^\lambda} A_\sigma + \psi^2 A_\nu \frac{\partial A_\sigma}{\partial x^\lambda} - \frac{\partial(\psi^2)}{\partial x^\sigma} A_\nu A_\lambda - \psi^2 \frac{\partial A_\nu}{\partial x^\sigma} A_\lambda - \psi^2 A_\nu \frac{\partial A_\lambda}{\partial x^\sigma} \right) \\ &= \frac{1}{2} \left(\frac{\partial(\psi^2 A_\lambda)}{\partial x^\nu} A^\mu + g^{\mu\sigma} \psi^2 A_\lambda \frac{\partial A_\sigma}{\partial x^\nu} + \frac{\partial(\psi^2 A_\nu)}{\partial x^\lambda} A^\mu + g^{\mu\sigma} \psi^2 A_\nu \frac{\partial A_\sigma}{\partial x^\lambda} - g^{\mu\sigma} \frac{\partial(\psi^2)}{\partial x^\sigma} A_\nu A_\lambda - g^{\mu\sigma} \psi^2 \frac{\partial A_\nu}{\partial x^\sigma} A_\lambda - g^{\mu\sigma} \psi^2 A_\nu \frac{\partial A_\lambda}{\partial x^\sigma} \right). \end{aligned}$$

The terms with A^μ cancel with the last term in the previous expansion of $\hat{\Gamma}_{\nu\lambda}^\mu$. This yields

$$\hat{\Gamma}_{\nu\lambda}^\mu = \Gamma_{\nu\lambda}^\mu + \frac{1}{2} g^{\mu\sigma} \left(\psi^2 A_\lambda \frac{\partial A_\sigma}{\partial x^\nu} + \psi^2 A_\nu \frac{\partial A_\sigma}{\partial x^\lambda} - \frac{\partial(\psi^2)}{\partial x^\sigma} A_\nu A_\lambda - \psi^2 \frac{\partial A_\nu}{\partial x^\sigma} A_\lambda - \psi^2 A_\nu \frac{\partial A_\lambda}{\partial x^\sigma} \right);$$

recalling that the electromagnetic tensor is defined by $F_{\mu\nu} = \frac{\partial}{\partial x^\mu} A_\nu - \frac{\partial}{\partial x^\nu} A_\mu$, we can simplify the above to

$$\hat{\Gamma}_{\nu\lambda}^\mu = \Gamma_{\nu\lambda}^\mu + \frac{\psi^2}{2} g^{\mu\sigma} (A_\lambda F_{\nu\sigma} + A_\nu F_{\lambda\sigma}) - \psi g^{\mu\sigma} \frac{\partial \psi}{\partial x^\sigma}.$$

We now examine the Christoffel symbols when the indices take the value 5; we note that the Christoffel symbols satisfy the identity $\hat{\Gamma}_{NL}^M = \hat{\Gamma}_{LN}^M$ so it suffices to look at the cases $\hat{\Gamma}_{\nu\lambda}^5$, $\hat{\Gamma}_{5\lambda}^\mu$, $\hat{\Gamma}_{5\lambda}^5$, $\hat{\Gamma}_{55}^\mu$, $\hat{\Gamma}_{55}^5$.

$$\begin{aligned} \hat{\Gamma}_{\nu\lambda}^5 &= \frac{1}{2} h^{5K} \left(\frac{\partial}{\partial x^\nu} h_{\lambda K} + \frac{\partial}{\partial x^\lambda} h_{\nu K} - \frac{\partial}{\partial x^K} h_{\nu\lambda} \right) \\ &= \frac{1}{2} h^{5\sigma} \left(\frac{\partial}{\partial x^\nu} h_{\lambda\sigma} + \frac{\partial}{\partial x^\lambda} h_{\nu\sigma} - \frac{\partial}{\partial x^\sigma} h_{\nu\lambda} \right) + \frac{1}{2} h^{55} \left(\frac{\partial}{\partial x^\nu} h_{\lambda 5} + \frac{\partial}{\partial x^\lambda} h_{\nu 5} - \frac{\partial}{\partial x^5} h_{\nu\lambda} \right) \\ &= -\frac{1}{2} A^\sigma \left(\frac{\partial}{\partial x^\nu} g_{\lambda\sigma} + \frac{\partial}{\partial x^\lambda} g_{\nu\sigma} - \frac{\partial}{\partial x^\sigma} g_{\nu\lambda} \right) - \frac{1}{2} A^\sigma \left(\frac{\partial}{\partial x^\nu} (\psi^2 A_\lambda A_\sigma) + \frac{\partial}{\partial x^\lambda} (\psi^2 A_\nu A_\sigma) - \frac{\partial}{\partial x^\sigma} (\psi^2 A_\nu A_\lambda) \right) \\ &\quad + \frac{1}{2} \left(A^\sigma A_\sigma + \frac{1}{\psi^2} \right) \left(\frac{\partial}{\partial x^\nu} (\psi^2 A_\lambda) + \frac{\partial}{\partial x^\lambda} (\psi^2 A_\nu) \right) \end{aligned}$$

We note that the first term is the Christoffel symbol $\Gamma_{\sigma\nu\lambda}$; we expand the second term as before with the Leibniz rule and get cancellation with the $A^\sigma A_\sigma$ terms. This leaves

$$\begin{aligned}\hat{\Gamma}_{\nu\lambda}^5 &= A^\sigma \Gamma_{\sigma\nu\lambda} - \frac{\psi^2}{2} A^\sigma \left(A_\lambda \frac{\partial A_\sigma}{\partial x^\nu} + A_\nu \frac{\partial A_\sigma}{\partial x^\lambda} - \frac{\partial A_\nu}{\partial x^\sigma} A_\lambda - A_\nu \frac{\partial A_\lambda}{\partial x^\sigma} \right) - \frac{1}{2} A^\sigma A_\nu A_\lambda \frac{\partial(\psi^2)}{\partial x^\sigma} + \frac{1}{2\psi^2} \left(\frac{\partial}{\partial x^\nu} (\psi^2 A_\lambda) + \frac{\partial}{\partial x^\lambda} (\psi^2 A_\nu) \right) \\ &= A^\sigma \Gamma_{\sigma\nu\lambda} - \frac{\psi^2}{2} A^\sigma (A_\lambda F_{\nu\sigma} + A_\nu F_{\lambda\sigma}) - \psi A^\sigma A_\nu A_\lambda \frac{\partial\psi}{\partial x^\sigma} + \frac{1}{\psi} \left(\frac{\partial\psi}{\partial x^\nu} A_\lambda + \frac{\partial\psi}{\partial x^\lambda} A_\nu \right) + \frac{1}{2} \left(\frac{\partial A_\lambda}{\partial x^\nu} + \frac{\partial A_\nu}{\partial x^\lambda} \right).\end{aligned}$$

For the next term we have

$$\begin{aligned}\hat{\Gamma}_{5\lambda}^\mu &= \frac{1}{2} h^{\mu K} \left(\frac{\partial}{\partial x^5} h_{\lambda K} + \frac{\partial}{\partial x^\lambda} h_{5K} - \frac{\partial}{\partial x^K} h_{5\lambda} \right) \\ &= \frac{1}{2} h^{\mu\sigma} \left(\frac{\partial}{\partial x^5} h_{\lambda\sigma} + \frac{\partial}{\partial x^\lambda} h_{5\sigma} - \frac{\partial}{\partial x^\sigma} h_{5\lambda} \right) + \frac{1}{2} h^{\mu 5} \left(\frac{\partial}{\partial x^5} h_{\lambda 5} + \frac{\partial}{\partial x^\lambda} h_{55} - \frac{\partial}{\partial x^5} h_{\nu\lambda} \right) \\ &= \frac{1}{2} g^{\mu\sigma} \left(\frac{\partial}{\partial x^\lambda} (\psi^2 A_\sigma) - \frac{\partial}{\partial x^\sigma} (\psi^2 A_\lambda) \right) - \frac{1}{2} A^\mu \left(\frac{\partial}{\partial x^\lambda} (\psi^2) \right) \\ &= \frac{1}{2} g^{\mu\sigma} \left(\psi^2 \frac{\partial A_\sigma}{\partial x^\lambda} - \psi^2 \frac{\partial A_\lambda}{\partial x^\sigma} - A_\lambda \frac{\partial(\psi^2)}{\partial x^\sigma} \right) \\ &= \psi g^{\mu\sigma} \left(\frac{\psi}{2} F_{\lambda\sigma} - A_\lambda \frac{\partial\psi}{\partial x^\sigma} \right)\end{aligned}$$

$$\begin{aligned}\hat{\Gamma}_{5\lambda}^5 &= \frac{1}{2} h^{5K} \left(\frac{\partial}{\partial x^5} h_{\lambda K} + \frac{\partial}{\partial x^\lambda} h_{5K} - \frac{\partial}{\partial x^K} h_{5\lambda} \right) \\ &= \frac{1}{2} h^{5\sigma} \left(\frac{\partial}{\partial x^5} h_{\lambda\sigma} + \frac{\partial}{\partial x^\lambda} h_{5\sigma} - \frac{\partial}{\partial x^\sigma} h_{5\lambda} \right) + \frac{1}{2} h^{55} \left(\frac{\partial}{\partial x^5} h_{\lambda 5} + \frac{\partial}{\partial x^\lambda} h_{55} - \frac{\partial}{\partial x^5} h_{\nu\lambda} \right) \\ &= -\frac{1}{2} A^\sigma \left(\frac{\partial}{\partial x^\lambda} (\psi^2 A_\sigma) - \frac{\partial}{\partial x^\sigma} (\psi^2 A_\lambda) \right) + \frac{1}{2} \left(A^\sigma A_\sigma + \frac{1}{\psi^2} \right) \left(\frac{\partial}{\partial x^\lambda} (\psi^2) \right) \\ &= -\frac{\psi^2}{2} A^\sigma \left(\frac{\partial A_\sigma}{\partial x^\lambda} - \frac{\partial A_\lambda}{\partial x^\sigma} \right) + \frac{1}{2} A^\sigma A_\lambda \frac{\partial(\psi^2)}{\partial x^\sigma} + \frac{1}{2\psi^2} \frac{\partial(\psi^2)}{\partial x^\lambda} \\ &= -\frac{\psi^2}{2} A^\sigma F_{\lambda\sigma} + \psi A^\sigma A_\lambda \frac{\partial\psi}{\partial x^\sigma} + \frac{1}{\psi} \frac{\partial\psi}{\partial x^\lambda}\end{aligned}$$

$$\begin{aligned}\hat{\Gamma}_{55}^\mu &= \frac{1}{2} h^{\mu K} \left(\frac{\partial}{\partial x^5} h_{5K} + \frac{\partial}{\partial x^5} h_{5K} - \frac{\partial}{\partial x^K} h_{55} \right) \\ &= \frac{1}{2} h^{\mu\sigma} \left(-\frac{\partial}{\partial x^\sigma} h_{55} \right) \\ &= -\psi g^{\mu\sigma} \frac{\partial\psi}{\partial x^\sigma}\end{aligned}$$

$$\begin{aligned}\hat{\Gamma}_{55}^5 &= \frac{1}{2} h^{5K} \left(\frac{\partial}{\partial x^5} h_{5K} + \frac{\partial}{\partial x^5} h_{5K} - \frac{\partial}{\partial x^K} h_{55} \right) \\ &= \frac{1}{2} h^{5\sigma} \left(-\frac{\partial}{\partial x^\sigma} h_{55} \right) \\ &= \psi A^\sigma \frac{\partial\psi}{\partial x^\sigma}\end{aligned}$$

With these we can compute the components of the Ricci curvature tensor $\hat{R}_{MN}$ via

$$\hat{R}_{MN} = \frac{\partial}{\partial x^L} \hat{\Gamma}_{MN}^L - \frac{\partial}{\partial x^N} \hat{\Gamma}_{ML}^L + \hat{\Gamma}_{LT}^L \hat{\Gamma}_{MN}^T - \hat{\Gamma}_{NT}^L \hat{\Gamma}_{ML}^T.$$

Moving forward, we assume for simplicity that $\psi = 1$.

The computation of the Ricci curvature tensor and the Ricci scalar are found in §17.5 of [1]. After a tedious computation we get

$$\begin{aligned}\widehat{R}_{\mu\nu} &= R_{\mu\nu} - \frac{1}{2}g_{\nu\sigma}F^{\lambda\sigma}F_{\lambda\mu} + \frac{1}{4}F_{\lambda\tau}F^{\lambda\tau}A_\nu A_\mu - \frac{1}{2}(A_\nu\nabla_\lambda(F_\mu^\lambda) + A_\mu\nabla_\lambda(F_\nu^\lambda))\widehat{R}_{5\nu} = -\frac{1}{2}\nabla_\lambda(F_\nu^\lambda) - \frac{1}{4}A_\nu F_\tau^\lambda F_\lambda^\tau \\ \widehat{R}_{55} &= \frac{1}{4}F_{\sigma\tau}F^{\sigma\tau}\end{aligned}$$

we write $F_\mu^\lambda = g_{\mu\sigma}F^{\lambda\sigma}$ and use the Leibniz rule along with the fact that $\nabla_\lambda(g_{\mu\sigma}) = 0$ to obtain $\nabla_\lambda(F_\mu^\lambda) = g_{\mu\sigma}\nabla_\lambda(F^{\lambda\sigma})$. Similarly, we can write

$$F_\tau^\lambda F_\lambda^\tau = g_{\sigma\tau}F^{\lambda\sigma}g^{\tau\rho}F_{\lambda\rho} = \delta_\sigma^\rho F^{\lambda\sigma}F_{\rho\lambda} = F^{\lambda\sigma}F_{\lambda\sigma}.$$

This allows us to rewrite the components of the Ricci curvature tensor as

$$\begin{aligned}\widehat{R}_{\mu\nu} &= R_{\mu\nu} - \frac{1}{2}g_{\nu\sigma}F^{\lambda\sigma}F_{\lambda\mu} + \frac{1}{4}F_{\lambda\tau}F^{\lambda\tau}A_\nu A_\mu - g_{\mu\sigma}A_\nu\nabla_\lambda(F^{\lambda\sigma}) \\ \widehat{R}_{\mu 5} &= -\frac{1}{2}g_{\mu\sigma}\nabla_\lambda(F^{\lambda\sigma}) - \frac{1}{4}A_\nu F^{\lambda\sigma}F_{\lambda\sigma} \\ \widehat{R}_{55} &= \frac{1}{4}F_{\sigma\tau}F^{\sigma\tau}.\end{aligned}$$

We then get the following form of the Ricci scalar, called the “Kaluza–Klein miracle”:

$$\widehat{R} = R - \frac{1}{4}F_{\sigma\tau}F^{\sigma\tau}.$$

3 The Einstein Field Equations

We use the form of the Ricci scalar obtained in the previous section to derive the vacuum Einstein field equations for the Kaluza–Klein metric. We take the equation

$$\widehat{R}_{MN} - \frac{1}{2}\widehat{R}h_{MN} = 0$$

and break it into cases dependent on whether M, N are 5. We begin with $M = N = 5$:

$$0 = \widehat{R}_{55} - \frac{1}{2}\widehat{R}h_{55} = \frac{1}{4}F_{\sigma\tau}F^{\sigma\tau} - \frac{1}{2}\left(R - \frac{1}{4}F_{\sigma\tau}F^{\sigma\tau}\right) = \frac{3}{8}F_{\sigma\tau}F^{\sigma\tau} - \frac{1}{2}R.$$

This gives that $R = \frac{3}{4}F_{\sigma\tau}F^{\sigma\tau}$. Now we examine the case where one of M, N is 5; we need only pick one of the two since the Ricci curvature tensor and metric are both symmetric.

$$0 = \widehat{R}_{\mu 5} - \frac{1}{2}\widehat{R}h_{\mu 5} = -\frac{1}{2}g_{\mu\sigma}\nabla_\lambda(F^{\lambda\sigma}) - \frac{1}{4}A_\mu F_{\lambda\tau}F^{\tau\lambda} - \frac{1}{2}RA_\mu;$$

using the previous identity for R we obtain

$$0 = -\frac{1}{2}g_{\mu\sigma}\nabla_\lambda(F^{\lambda\sigma}) - \frac{1}{2}A_\mu F_{\tau\lambda}F^{\tau\lambda}.$$

We contract with the dual metric $g^{\mu\sigma}$ and rearrange to obtain

$$\nabla_\lambda(F^{\lambda\sigma}) = -\frac{1}{4}A^\sigma F_{\tau\lambda}F^{\tau\lambda}.$$

This may be thought of as a generalization of Maxwell's equations $\frac{\partial}{\partial x^\mu} F^{\mu\sigma} = J^\sigma$ to the Kaluza–Klein spacetime.

Finally, we examine $M = \mu$ and $N = \nu$:

$$\begin{aligned} 0 &= \widehat{R}_{\mu\nu} - \frac{1}{2}\widehat{R}h_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\nu\sigma}F^{\lambda\sigma}F_{\lambda\mu} + \frac{1}{4}F_{\lambda\tau}F^{\lambda\tau}A_\nu A_\mu - g_{\mu\sigma}A_\nu\nabla_\lambda(F^{\lambda\sigma}) - \frac{1}{2}\left(R - \frac{1}{4}F_{\sigma\tau}F^{\sigma\tau}\right)(g_{\mu\nu} + A_\mu A_\nu) \\ &= R_{\mu\nu} - \frac{1}{2}g_{\nu\sigma}F^{\lambda\sigma}F_{\lambda\mu} + \frac{1}{4}F_{\lambda\tau}F^{\lambda\tau}A_\nu A_\mu - \frac{1}{4}g_{\mu\nu}F_{\sigma\tau}F^{\sigma\tau} - \frac{1}{4}F_{\sigma\tau}F^{\sigma\tau}A_\mu A_\nu + A_\nu A_\mu F_{\tau\lambda}F^{\tau\lambda} \\ &= R_{\mu\nu} - \frac{1}{2}g_{\nu\sigma}F^{\lambda\sigma}F_{\lambda\mu} - \frac{1}{4}g_{\mu\nu}F_{\sigma\tau}F^{\sigma\tau} + A_\nu A_\mu F_{\tau\lambda}F^{\tau\lambda}. \end{aligned}$$

This gives

$$R_{\mu\nu} = \frac{1}{4}g_{\mu\nu}F_{\sigma\tau}F^{\sigma\tau} + \frac{1}{2}F_\nu^\lambda F_{\lambda\mu} + A_\mu A_\nu F_{\sigma\tau}F^{\sigma\tau}.$$

Contracting with the dual metric, we get

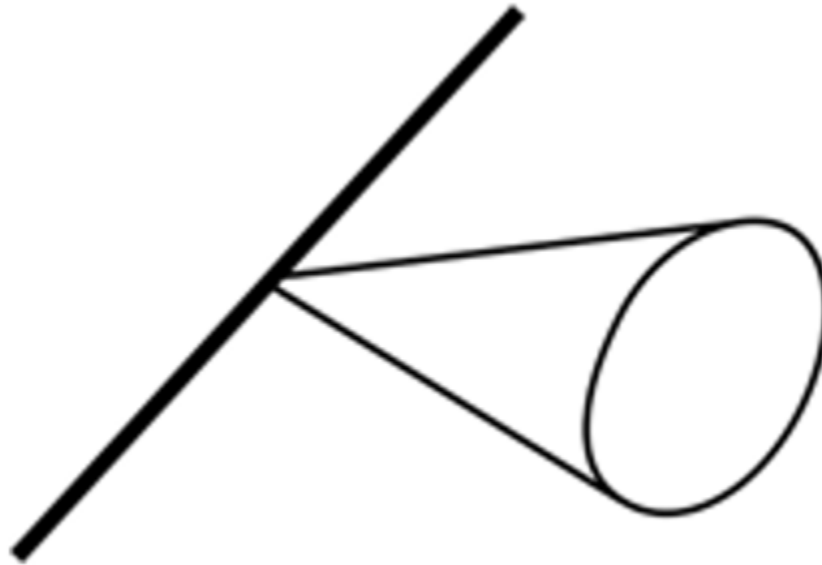
$$R = F_{\sigma\tau}F^{\sigma\tau} + \frac{1}{2}F^{\lambda\mu}F_{\lambda\mu} + A^\nu A_\nu F_{\sigma\tau}F^{\sigma\tau};$$

using the expression $R = \frac{3}{4}F_{\sigma\tau}F^{\sigma\tau}$ we obtain

$$F_{\sigma\tau}F^{\sigma\tau}\left(\frac{3}{4} + A^\nu A_\nu\right) = 0.$$

4 The Fifth Dimension

It is unclear what the fifth dimension should be physically; does it even exist? Oskar Klein posited that this dimension does, in fact exist; however it has been unobserved since it is “compactified” into a small circle. This is best described with the following image:



Here the cylinder looks like a line when observed from far, but when observed closely we see that there is another dimension which makes the “line” into a cylinder.

References

- [1] Ta-Pei Cheng. *Einstein's Physics: Atoms, Quanta, and Relativity - Derived, Explained, and Appraised*. Oxford, 2013.